NovaTech Workforce Attrition Analysis

Excel End-to-End Portfolio Case Study

Project Overview

Employee turnover can create recruitment costs, operational disruption, lost institutional knowledge, and reduced team performance. NovaTech Solutions required an analytical review of its workforce data to identify where attrition was concentrated and which employee characteristics or workplace conditions were associated with a greater likelihood of departure.

I completed an end-to-end HR analytics project in Microsoft Excel, beginning with a raw employee dataset and finishing with an interactive executive dashboard, documented business insights, and evidence-based recommendations.

Business Objective

The primary objective was to answer the following questions:

- What is the organization's overall attrition rate?
- Which departments and job roles have the highest attrition?
- Is overtime associated with employee turnover?
- Which age and tenure groups have the greatest retention risk?
- Does job satisfaction appear to be associated with attrition?
- Does income meaningfully differentiate attrition risk?
- Are gender or marital status important turnover indicators?
- What actions should HR leadership prioritize?

Dataset

The raw dataset contained 1,488 employee records and 35 original fields. It included information relating to:

- Employee demographics
- Department and job role
- Monthly income
- Overtime
- Job satisfaction
- Years at the company
- Attrition status
- Education
- Marital status
- Performance and employment characteristics

Following cleaning and duplicate handling, the final analytical dataset contained 1,468 unique employee records. The dataset used for this project is synthetic and does not contain real employee information.

Tools Used

- Microsoft Excel
- Excel Tables
- Structured references
- Data-cleaning formulas
- PivotTables
- PivotCharts
- Slicers and dashboard controls
- Conditional formatting
- KPI calculations
- Business analysis and reporting

Workbook Structure

The final workbook was separated into seven layers:

1. **Raw_Data** — original source data preserved for auditability
2. **Cleaned_Data** — standardized and analysis-ready employee data
3. **Data_Dictionary** — definitions of fields and calculated columns
4. **Pivot_Tables** — analytical summaries supporting dashboard visuals
5. **Dashboard** — interactive executive attrition dashboard
6. **Insights** — quantified analytical findings
7. **Recommendations** — business actions connected to the findings

This structure separates source data, transformation, analysis, presentation, and interpretation.

Data-Cleaning Process

Duplicate handling

I reviewed employee identifiers and removed duplicated employee records before analysis. The clean dataset retained one valid record per employee.

Missing department values

Nineteen employees did not have a recorded department. Rather than deleting these employees or assigning them to an unsupported department, I retained them under the category **Unknown Department**.

This decision preserved the employee population while making the limitation visible in the analysis.

Text standardization

Categorical fields were reviewed for:

- Extra spaces
- Capitalization differences
- Inconsistent labels
- Missing or unknown values
- Department-name variations

Values were standardized to ensure that equivalent categories were grouped correctly in PivotTables.

Date and numerical validation

Date and numerical fields were checked to ensure they could be used correctly in calculations, grouping, and analysis.

Calculated analytical fields

Additional fields were created to improve the business analysis, including:

- Age Group
- Tenure Group
- Income Group
- Attrition indicator or count field
- Clean department classification

These fields allowed management-friendly comparisons rather than relying only on raw numeric values.

Analysis and Dashboard Design

The dashboard was designed as an executive decision-support tool rather than a collection of unrelated charts.

KPI section

The dashboard highlights:

- Total employees: 1,468
- Departures: 226
- Active employees: 1,242
- Attrition rate: 15.4%
- Average employee age
- Average monthly income
- Average tenure

Attrition segmentation

Attrition was analyzed by:

- Department
- Job role
- Overtime status
- Age group
- Tenure group
- Job satisfaction
- Income group
- Gender
- Marital status

Both attrition rate and employee volume were considered. This prevented small groups with unusually high percentages from being incorrectly presented as the company's largest business problem.

Key Findings

Overall attrition

The company recorded an attrition rate of 15.4%, representing 226 departures among 1,468 employees.

Sales was the highest-risk department

Sales recorded a 19.6% attrition rate, with 87 departures among 443 employees. This was higher than both Human Resources and Research and Development.

Because Sales had both a high attrition rate and a large employee population, it represented a meaningful retention priority.

Job-role risk required sample-size interpretation

HR Business Partner recorded the highest job-role attrition rate at 23.8%, but the result was based on only 21 employees and five departures.

Sales Representative recorded a slightly lower rate of 21.7%, but the role had 152 employees and 33 departures. It was therefore the more operationally significant job-role concern.

Overtime showed the strongest observed association

Employees working overtime had a 23.6% attrition rate, compared with 12.3% among employees who did not work overtime.

This was the largest observed difference across the principal variables examined.

Attrition was concentrated among younger employees

Employees under age 25 recorded a 26.3% attrition rate. Attrition generally declined among older employee groups.

This finding suggests that early-career support and development may be important retention factors.

First-year retention was a major concern

Employees with less than one year of tenure had an attrition rate of 21.3%. The rate declined to 10.8% among employees with ten or more years at the company.

This pattern indicates a potential onboarding, recruitment-fit, or early-engagement gap.

Satisfaction was associated with lower attrition

Employees reporting low job satisfaction had a 20.2% attrition rate, compared with 12.9% among employees reporting very high satisfaction.

The overall pattern showed declining attrition as satisfaction improved.

Income findings required caution

The Low Income category recorded a 29.4% attrition rate, but it contained only 17 employees. Because of the small sample, this finding was treated as a hypothesis for monitoring rather than a confirmed company-wide pattern.

Gender and marital status were weak differentiators

Attrition rates across gender and marital-status groups were relatively close. These variables did not appear to be major attrition indicators in the current dataset.

Business Recommendations

Based on the analysis, I recommended that NovaTech:

1. Review overtime, staffing levels, and workload distribution, especially within Sales.
2. Introduce structured 30-, 60-, and 90-day onboarding check-ins.
3. Investigate Sales Representative workload, incentives, targets, and management support.
4. Review the HR Business Partner population while recognizing its small sample size.
5. Connect satisfaction surveys to regular manager and career-development conversations.
6. Monitor compensation patterns using larger samples and role-adjusted salary analysis.
7. Refresh the dashboard monthly or quarterly when live HR data becomes available.

Analytical Limitations

The dashboard is based on descriptive analysis. It demonstrates associations but cannot prove that one variable caused attrition.

Other limitations include:

- Synthetic rather than real company data
- Missing department values for 19 employees
- Small populations in certain analytical categories
- No exit-interview information
- No direct manager or team-level variables
- No longitudinal salary or promotion history
- No external compensation benchmark

Skills Demonstrated

This project demonstrates my ability to:

- Clean and standardize raw business data
- Preserve an auditable workbook structure
- Create calculated analytical categories
- Build PivotTables and PivotCharts
- Design executive KPI dashboards
- Evaluate rates together with sample sizes
- Avoid over-interpreting small groups
- Translate analysis into business recommendations
- Explain data limitations transparently
- Communicate findings to both technical and nontechnical stakeholders

Final Outcome

The project transformed a raw HR dataset into a complete analytical product consisting of:

- Cleaned employee-level data
- A documented data dictionary
- Reusable PivotTable analysis
- An interactive Excel dashboard
- Quantified business insights
- Action-oriented HR recommendations
- An executive reporting package

The final result provides HR leadership with a structured view of where attrition is concentrated and which areas should be investigated first.